

1. What Paper 3 is actually about

From the meeting, there are really two intertwined ideas, and I think they should be framed as one coherent contribution rather than two separate add-ons:

Core idea: Replace the hand-crafted, shape-classifier-driven extrapolation (Paper 1) with a learned extrapolation model trained on large-scale simulation data (using Berith's Isaac Gym/Abaqus pipeline). Instead of "classify as cuboid/sphere/cylinder → apply shape-specific rule," you'd have a network that takes the initial 7×8 contact and directly predicts the extended contact manifold, learned from thousands of simulated grasps across arbitrary geometries.

Secondary idea: Extend the regrasp optimizer from Paper 2 with a third cost term — a task-oriented objective (Vincent's "task-oriented graph/objective" point) — and re-validate the whole regrasp pipeline at scale using the new simulation-based data instead of the 86-configuration physical dataset.

These two ideas are connected by a shared narrative: Papers 1 and 2 showed the *concept* works (extrapolation is geometrically/predictively sound; CDT is the best search strategy), but both were bottlenecked by small physical datasets (650 points, 602 trials) and a brittle 3-way shape classifier. Paper 3's contribution is *scaling up* via simulation — both for training a better extrapolation model and for validating the regrasp framework across far more geometries than 15 physical objects.

2. What you can reuse from Papers 1 & 2 and Berith's work

From Paper 1 (extrapolation):

- The evaluation metrics are gold and should carry over almost unchanged: Tactile Centroid (TC) error, GSR-based error, SSIM, and the "Safe Zone" concept. These give you a ready-made validation framework for the *new* learned extrapolation model — you can directly compare "shape-classifier + heuristic rules" (old) vs. "learned extrapolation network" (new) on the same metrics.
- The GSR network from Kwiatkowski et al. stays as the stability evaluator — no need to retrain that.
- The grid-scan data collection protocol (Fig. 1e/1j in Paper 1) is your template for *real* validation trials, but now Isaac Gym lets you do this at massive scale virtually.

From Paper 2 (regrasp optimization):

- The entire digital twin / virtual search architecture (TIS objective, dual-objective cost function, CDT paradigm, reachability filtering) is reusable wholesale. CDT was the winner — that's your default search mode going forward.
- The 602-trial evaluation framework (Δ GSR, robustness matrix, fidelity error, Pareto frontier) is your template for re-validating at scale.
- The third cost term Vincent mentioned (task-oriented objective) slots into $J(P)$ as a new λ -weighted term: $J(P) = \lambda_q \cdot C_{\text{quality}} + \lambda_e \cdot C_{\text{effort}} + \lambda_t \cdot C_{\text{task}}$.

From Berith's paper (simulation infrastructure):

- The Isaac Gym (elastic, fast) / Abaqus (hyperelastic, slow, more accurate) hybrid pipeline, with the selection criteria (NPt/CHt, NPo/CHo ratios) for choosing which to use.
- The CNN that maps deformation maps → synthetic 28-tixel tactile output (this is your "virtual sensor" — critical, since this is what lets you generate realistic tactile imprints from arbitrary STL geometries).
- Important caveat from her paper: Isaac Gym averaged SSIM 0.63 vs Abaqus 0.72 against real data, and Isaac Gym specifically struggles with detailed/relief geometries (missing entire features, per Fig. 7 objects 5–6). For your purposes — generating *many* contact-extrapolation training samples around simple-ish object regions — Isaac Gym's speed (37s avg vs 162s for Abaqus) probably wins for bulk data generation, but you may want Abaqus for validation subsets or for objects with fine features, following her selection criteria.

3. Starting point for working with the simulation

Before touching Berith's GitHub code, I'd want to nail down the experimental design, because it determines what you actually need from the simulator:

Step 1 — Define the data generation protocol (this is the "snapshot grid" idea from the meeting). For a given object (STL mesh) and a given initial grasp pose, generate a grid of nearby contact poses (e.g., $\pm 10\text{mm}$ in x/y, small rotations) — exactly like Paper 1's grid-scan, but virtual and at much higher density (hundreds instead of tens per object). Each grid point produces a synthetic 7×8 (or 7×4 per sensor) tactile map via Berith's CNN pipeline.

Step 2 — Define the learning task precisely. Input: initial tactile imprint (single snapshot, or — per Vincent's idea — a short sequence of 3-5 snapshots during closing). Output: extended/extrapolated tactile manifold (M_{ext}) at higher resolution, the same object Paper 1 was solving with shape-classification + heuristics. This is a direct image-to-image (or sequence-to-image) regression/generation problem — likely a CNN or small U-Net, replacing the classifier+rules pipeline.

Step 3 — Practical first milestone with Berith's code: get a single object's STL into the pipeline, generate one grasp, then programmatically perturb the gripper pose on a grid and confirm you can batch-generate tactile maps without manual intervention. This is the make-or-break engineering step — if Isaac Gym generation is fast and stable enough for batch automation (her Table 3 numbers suggest yes, $\sim 37\text{s}$ average), the rest follows. I'd want to see her code structure (how the scene/objects/gripper are scripted) before planning further — yes, please share it when ready.

Step 4 — Decide on the multi-snapshot question early, since it changes the network architecture (single-frame CNN vs. something that takes a short temporal sequence, e.g., a small 3D-CNN or RNN/transformer over 3-5 frames). Vincent seemed enthusiastic about this; it's a

nice point of novelty over Paper 1 but adds complexity. I'd treat it as a stretch goal / ablation rather than a hard requirement for the first pass.

4. My overall take

This is a sensible, well-scoped third paper that closes the loop nicely: Paper 1 validated extrapolation fidelity with a heuristic approach on a small dataset; Paper 2 validated the regrasp search on top of that; Paper 3 says "now let's learn the extrapolation directly from large-scale simulation, removing the shape-classification dependency entirely (which conveniently also makes the classifier-robustness criticism from Paper 2 moot — there's no classifier anymore), and re-validate the full regrasp pipeline at a scale physical trials can't reach."

A few things I'd flag as you plan:

- **Scope risk:** the meeting touched on at least three extensions (simulation-based extrapolation, multi-snapshot input, task-oriented cost term). Doing all three well for one paper is ambitious. I'd suggest the simulation-based extrapolation network is the spine of the paper, with the task-oriented cost term as a secondary contribution (it's a relatively small addition to existing machinery), and multi-snapshot as an ablation/future-work item unless time allows.
- **The sim-to-real gap is now load-bearing in a new way.** Papers 1/2 validated against real hardware. If Paper 3's extrapolation model is trained *and* validated mostly in simulation, reviewers will ask how well it transfers — Berith's SSIM numbers (0.63-0.72 average, much worse on detailed objects) suggest this gap is real and non-trivial. I'd budget for at least a modest real-world validation set (even 5-10 objects, similar to Paper 1's external set) to show sim-trained extrapolation still works on real tactile data.
- **Framing vs. Paper 1:** be explicit about what's "thrown away" (the 3-way shape classifier and hand-coded rules) vs. what's preserved (the evaluation metrics, the GSR network, the overall extrapolate→search→regrasp pipeline). This makes the contribution legible as "Paper 1's concept, now learned and scaled" rather than a totally different system.